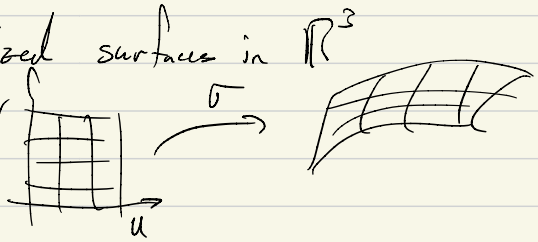
Introduction

Manifolds:

1. In calculus we study parametrized surfaces in $\mathbb{R}^3$

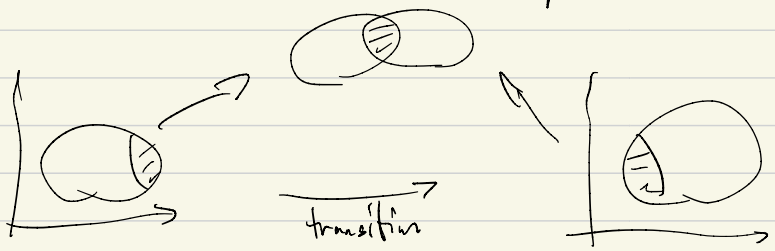
$$\sigma : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3$$



σ is <u>regular</u> if σ_u and σ_v are linearly indep.
$(\sigma_u \times \sigma_v \neq 0)$

2. Some surfaces, such as the unit sphere S^2 require multiple maps to parametrize

Inverse fn. thm: transition maps are smooth



3. <u>manifold</u> = a topological space (with some restrictions) parametrized by a set of coordinate patches from $\mathbb{R}^d$ with continuous transitions

<u>smooth manifold</u>: smooth transition maps

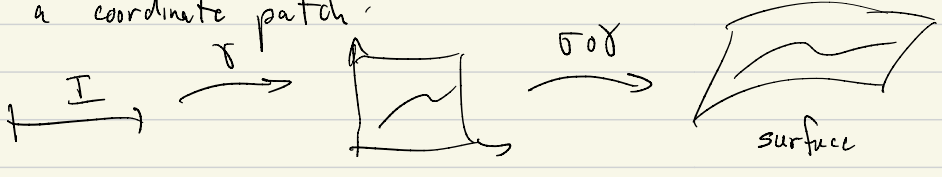
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Riemannian metrics:

1. length of a parametrized curve $\gamma : I \longrightarrow \mathbb{R}^3$

$$l(\gamma) = \int_I \|\dot{\gamma}\| dt \qquad \dot{\gamma}(t) := \frac{d}{dt}\gamma(t)$$

In a coordinate patch:



Let $\gamma(t) = (u(t), v(t))$

$$\frac{d}{dt}(\sigma \circ \gamma) = \sigma_u \dot{u} + \sigma_v \dot{v}$$

$$\| \quad \|^2 = \sigma_u^2 \dot{u}^2 + 2(\sigma_u \cdot \sigma_v)\dot{u}\dot{v} + \sigma_v^2 \dot{v}^2$$

$$= g(\dot{\gamma}, \dot{\gamma}) \quad \leftarrow \text{quadratic form (depends on } u, v\text{)} \qquad \begin{pmatrix} \sigma_u^2 & \sigma_u \sigma_v \\ \sigma_u \sigma_v & \sigma_v^2 \end{pmatrix}$$

$$\ell(\gamma) = \int_I \sqrt{g(\dot{\gamma}, \dot{\gamma})}\, dt$$

For surfaces in $\mathbb{R}^3$, g is called the <u>first fundamental form</u>

Think of this as a legend for the map

3. <u>Riemannian metric</u> = a smoothly varying quadratic form defined on the tangent space to a manifold

This gives an infinitesimal measure of distance

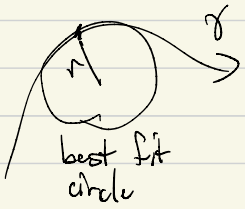
ds^2 = quadratic form

$\Rightarrow$ length $\ell = \int ds$

The metric also determines angles, area, volume, etc.

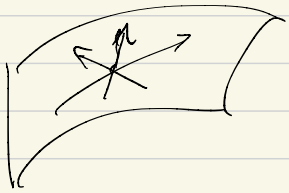
<u>Curvature</u>

1. For a curve in $\mathbb{R}^3$, curvature is defined by radius



$K = 1/r$

2. On a surface we can consider a family of curves through a given pt.



The <u>second fundamental form</u> II computes these local curvatures

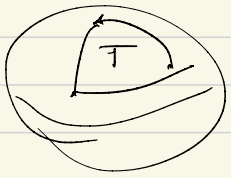
3. Gauss curvature $K = \det(\mathrm{II})$

Theorema Egregium: K depends only on g (and not on II!)

Furthermore, K measures the departure from Euclidean geometry

$$K(p) = \lim_{T \to p} \frac{\text{angle sum}(T) - \pi}{\text{area}(T)}$$

example: on the unit sphere, $\text{angle sum}(T) = \pi + \text{area}(T)$ $(K = 1)$



4. Riemannian curvature

Riemann showed that $\exists$ "geodesic" coordinates for which

$$g = \underset{\text{(identity matrix)}}{\text{Euclidean}} + \text{quadratic correction} + O(r^3)$$

The quadratic correction defines a curvature tensor R